Retail Business Performance & Profitability Analysis

Abstract

This project analyzes transactional retail data from the Sample Superstore dataset (9,994 order line-items, 2014–2017) to identify profit-draining product categories, evaluate the relationship between shipping/inventory lag and profitability, and surface actionable insights for category, regional, and discount strategy. The analysis combines SQL-based aggregation, Python (Pandas/Seaborn) exploratory analysis and correlation testing, and an interactive dashboard to communicate findings to non-technical stakeholders.

Introduction

Retail businesses generate large volumes of transactional data, but high sales volume does not always translate into healthy profit. The objective of this project is to move beyond top-line revenue and examine profitability at the category, sub-category, region, and discount level, in order to recommend where the business should invest, discount more cautiously, or discontinue underperforming product lines.

Tools Used

Layer	Tool / Library	Purpose
Data storage & cleaning	SQL (SQLite)	Null/duplicate checks, profit-margin aggregation queries
Analysis	Python – Pandas, NumPy	Data cleaning, feature engineering, correlation analysis
Visualization (static)	Matplotlib, Seaborn	EDA charts: margin by sub-category, correlation heatmap
Dashboard (interactive)	Plotly (HTML)	Filterable, browser-based dashboard for region/category/season
Version control	Git & GitHub	Project hosting and documentation

Steps Involved in Building the Project

- Imported the Superstore CSV (9,994 rows, 21 columns) into SQL; validated zero null values and zero duplicate line-items.
- Wrote SQL queries to compute profit margin (%) by Category and Sub-Category, and to rank regions and states by profitability.
- Used Python/Pandas to engineer features: Profit Margin, Shipping Days (Ship Date – Order Date, used as an inventory-lag proxy), Order Month and Season.
- Ran Pearson correlation between Shipping Days, Discount, Sales and Profit to test whether slower fulfillment relates to lower profitability.
- Built Seaborn/Matplotlib charts (margin by sub-category, correlation heatmap, regional and seasonal trends, discount-vs-margin scatter).
- Built an interactive Plotly HTML dashboard with KPI cards and filterable region/category/time views as a portable Tableau-style deliverable.
- Synthesized the findings below into category, regional and discount recommendations.

Key Findings

- **Furniture is the profitability problem child**: it earns nearly as much revenue as Technology ($742K vs $836K) but only a 2.5% profit margin, versus 17.4% for Technology and 17.0% for Office Supplies.
- **Tables and Bookcases are losing money outright**: Tables lose $17,725 on $206,966 of sales (–8.6% margin); Bookcases lose $3,473 (–3.0% margin); Office Supplies > Supplies also loses money (–2.5% margin).
- **Discount is the dominant driver of lost profit, not shipping delay.** Discount correlates strongly negatively with profit margin (r = –0.86), while shipping/inventory lag shows virtually no relationship with margin (r = –0.01) — slow fulfillment is not what is eroding profit; aggressive discounting is.

- **The West and East regions are the most profitable** (14.9% and 13.5% margin respectively), while Central trails at 7.9% margin despite solid sales volume — a regional discounting or cost issue worth investigating.
- **Sales and profit do not move in lockstep** (r = 0.48 between Sales and Profit), confirming that revenue alone is a poor proxy for business health on this dataset.

Conclusion

The analysis shows that profitability problems in this dataset are concentrated, not diffuse: two sub-categories (Tables, Bookcases) and a discounting pattern — not shipping speed — explain most of the margin erosion. Recommended actions are to (1) cap or restructure discounts on Tables, Bookcases and Supplies, where heavy discounting pushes margin negative, (2) reassess Furniture pricing and supplier costs given its 2.5% margin versus 17%+ for other categories, and (3) review Central-region pricing/discount policy to close the gap with West and East. These findings, the SQL queries, Python analysis, and interactive dashboard together form a reusable template for ongoing profitability monitoring.

Dataset: Sample Superstore (Kaggle) · Full code, SQL queries and dashboard: see project GitHub repository.